Lab # 01

COURSE CODE: SE-206L

COURSE TITLE:

Software Construction & Development

LAB TITLE

Environment Setup, Version Control, and Introduction to Java

LAB ASSESSMENT TABLE:

Obtained Marks	Total Marks

Submitted To: Engr. Rizwan Shah.

Submitted By: Muhammad Danial.

Registration – No: 24ABSWE0021.

Semester: 5th

Date: 20-09-2026.

Dept. of Software Engineering

UET Peshawar, Abbottabad Campus.

Instructor Sign : _____

Lab 1:

Environment Setup, Version Control, and Introduction to Java

Objective:

By the end of this lab, you will have a fully functioning Java development environment, a configured version control system, and your first working Software Configuration Item (SCI) hosted on GitHub.

Task 1: Install the Java Development Kit (JDK)

To write and run Java code, you need the JDK. It contains the Java compiler (javac for **Build time**) and the Java Virtual Machine (JVM for **Run-time**).

Instructions:

1. Go to the official Oracle website or Adoptium (Eclipse Temurin) and download the latest Long-Term Support (LTS) version of the JDK (e.g., JDK 17 or 21) for your operating system.
2. Run the installer and follow the default prompts.

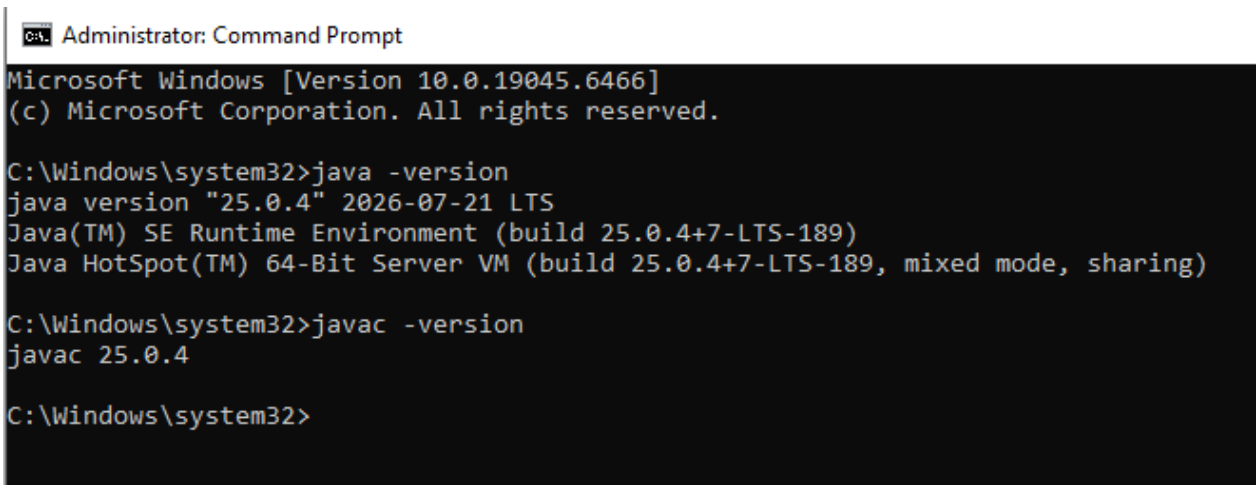
3. Configure Environment Variables (Windows):

- Open the Start Menu and search for "Environment Variables".
- Under System Variables, click New and create a variable named JAVA_HOME. Set the value to your JDK installation path (e.g., C:\Program Files\Java\jdk 21).
- Find the Path variable, click Edit, and add a new entry: %JAVA_HOME%\bin.

4. Verify Installation:

- Open a Command Prompt or Terminal.
- Type java -version and press Enter. You should see the Java Run-time version.
- Type javac -version and press Enter. You should see the Java Compiler version.

Solution # 1 Screenshot:



Task 2: Install the IDE (Apache NetBeans)

An Integrated Development Environment (IDE) automates the build process and organizes your workspace.

Instructions:

1. Navigate to the Apache NetBeans download page (netbeans.apache.org).
2. Download the latest installer for your operating system. (Note: NetBeans requires the JDK to be installed first, which is why we did Task 1).
3. Run the installer with the default settings.
4. Open Apache NetBeans to ensure it launches successfully.

Task 3: Create a GitHub Account & Install Git

In Software Construction, tracking the history of your code (Code Churn and Versioning) is mandatory. We use Git for local version control and GitHub for cloud hosting.

Instructions:

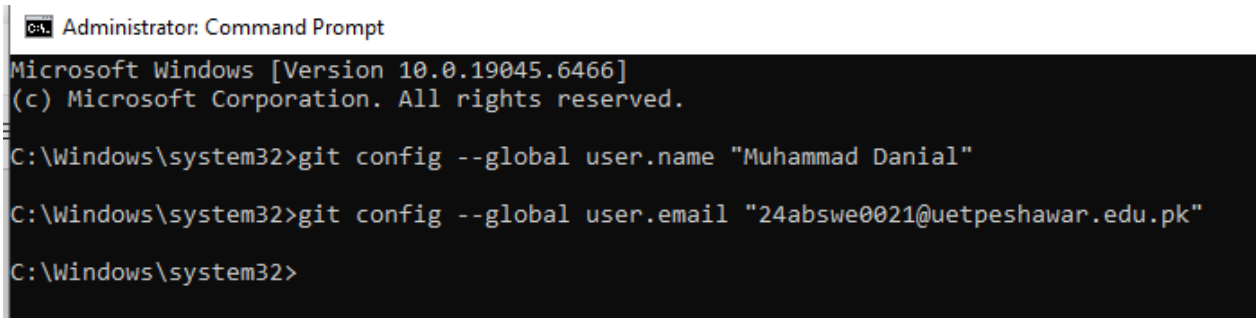
1. **Create an Account**: Go to github.com and sign up. It is highly recommended to use your university email address so you can access the GitHub Student Developer Pack later.

2. **Install Local Git:**
    - Go to git-scm.com and download the Git installer for your OS.
    - Run the installer using all the default settings (just keep clicking "Next").

3. **Configure Git:**

- Open your Command Prompt or Terminal.
- Tell Git who you are by running these two commands (use your actual name and GitHub email):
    - git config --global user.name "Your First and Last Name"
    - git config --global user.email "your.email@example.com"

Solution # 3 Screenshot:

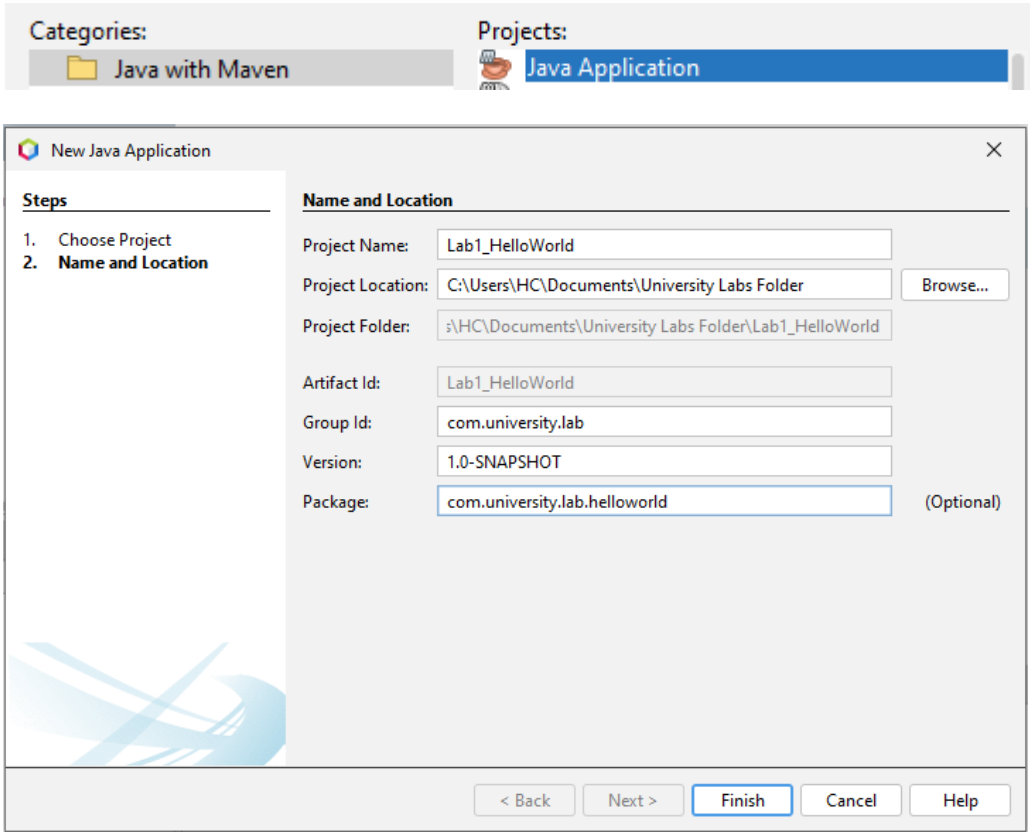


Task 4: Create a Project in Java

We will now use NetBeans to generate a standard Java project structure using Maven.
Instructions:

1. Open NetBeans and go to File > New Project.
2. Under the Categories pane on the left, select Java with Maven. Maven is our Build Automation Tool.
3. Under the Projects pane on the right, select Java Application. Click Next.
4. Configure your project:
    - Project Name: Lab1_HelloWorld
    - Project Location: Choose a dedicated folder for your university labs.
    - Group Id: com.university.lab
    - Package: com.university.lab.helloworld
5. Click Finish. NetBeans will generate the src and target folders discussed in the lecture.

Solution # 4 Screenshot:



Task 5: Write and Execute the Code

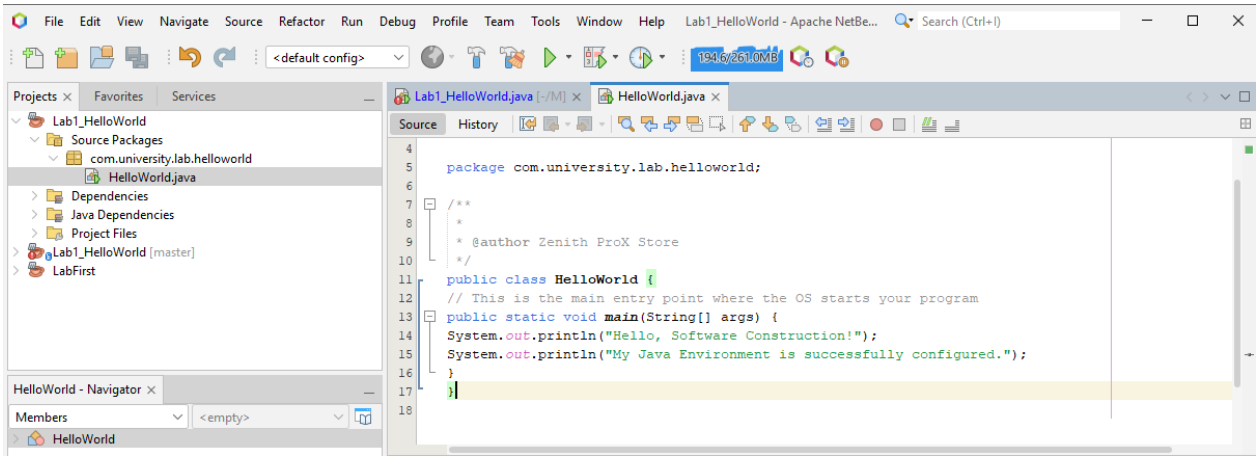
Let's create your first Java source code file.

Instructions:

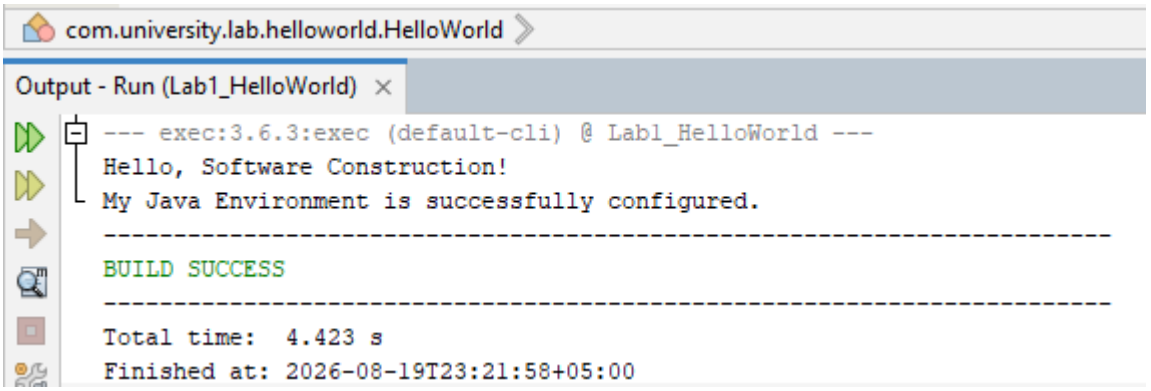
1. In NetBeans, look at the **Projects** window on the left. Expand Lab1_HelloWorld > Source Packages > com.university.lab.helloworld.
2. Double-click the generated .java file, or right-click the package and choose New > Java **Class** if one wasn't created. Name it HelloWorld.
3. Type the following code exactly as it appears. Notice that unlike C++ or Python, all Java code must exist inside a Class.

```
package com.university.lab.helloworld;
public class HelloWorld {
// This is the main entry point where the OS starts your program
public static void main(String[] args) {
System.out.println("Hello, Software Construction!");
System.out.println("My Java Environment is configured.");
 }
}
```

4. **Run the Code:** Click the green "Play" button in the top toolbar. You should see the text print out in the output console at the bottom of the screen.



Output:

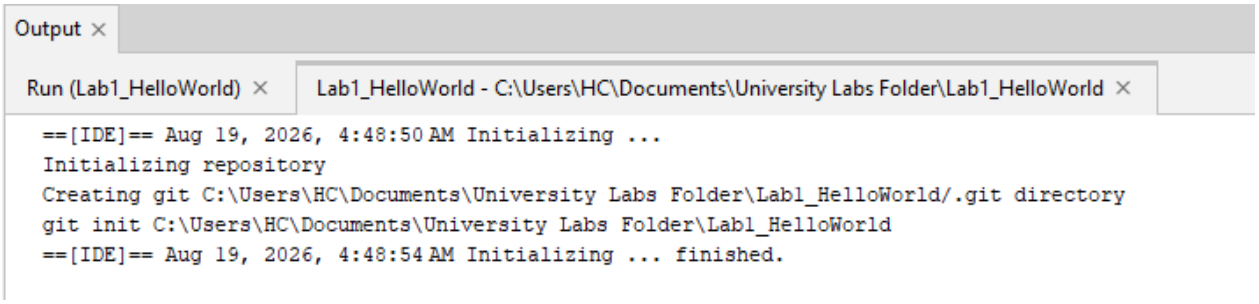


Task 6: Version Control Your Project

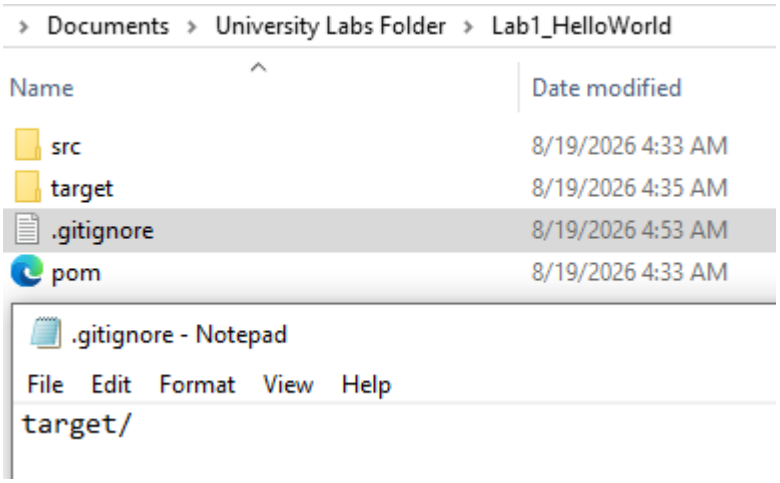
Your code is currently a local Software Configuration Item (SCI). We need to commit it to version control and push it to GitHub.

Instructions:

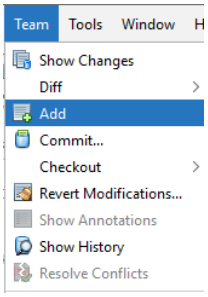
1. Log into GitHub on your web browser. Click the + icon in the top right and select **New repository.**
2. Name the repository Software-Construction-Labs. Leave it Public and click **Create repository.**
3. In NetBeans, go to the top menu bar and select **Team > Git > Initialize Repository**. Select your project's root folder.



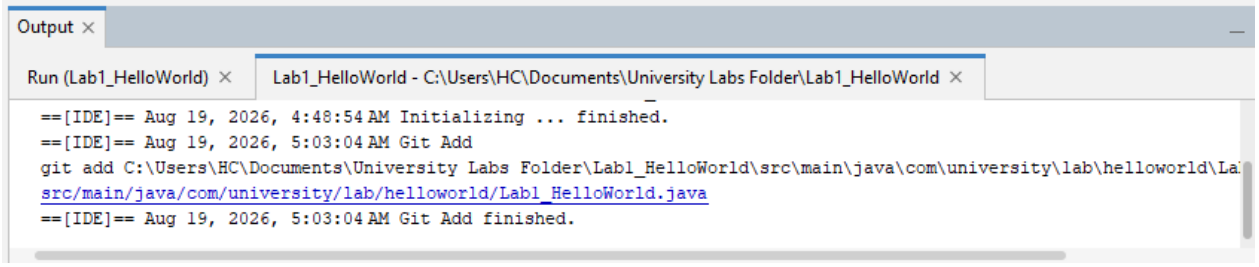
4. **CRITICAL STEP - The .gitignore File:**
   - In your project folder, create a plain text file named .gitignore.
   - Add the word target/ to this file and save it. This prevents Git from tracking your compiled run-time binaries. We only version control build-time source code.



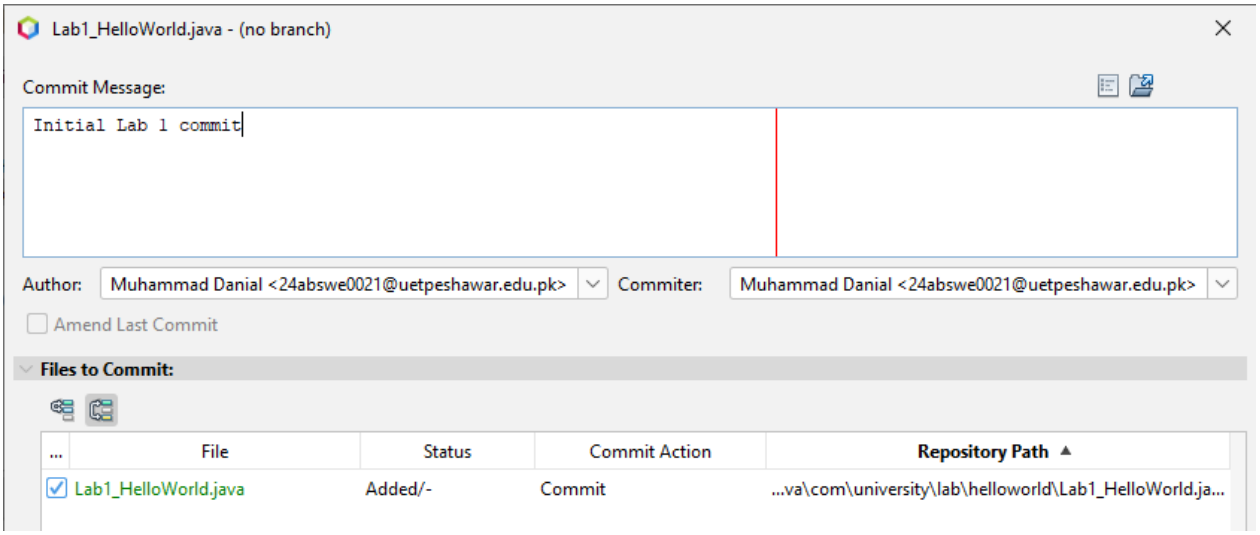
5. In NetBeans, right-click your project name, go to Git > Add.



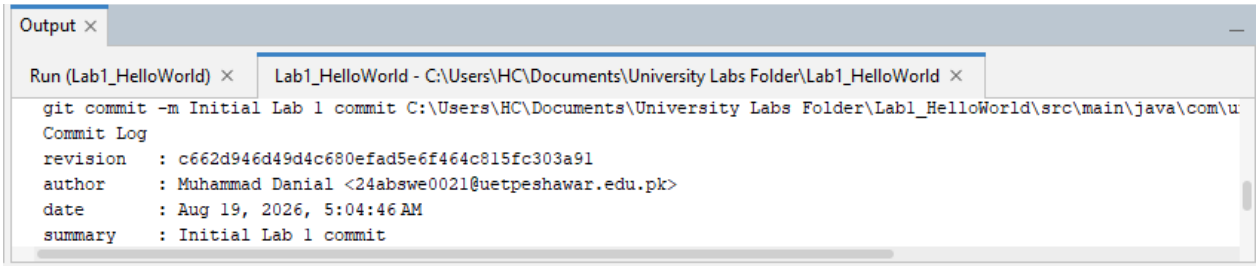
Output



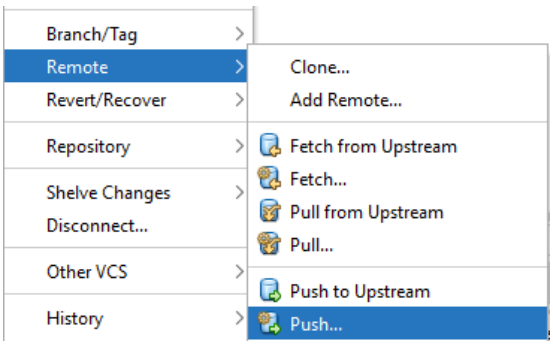
6. Right-click your project name again, go to Git > Commit. Type "Initial Lab 1 commit" in the message box and click Commit.

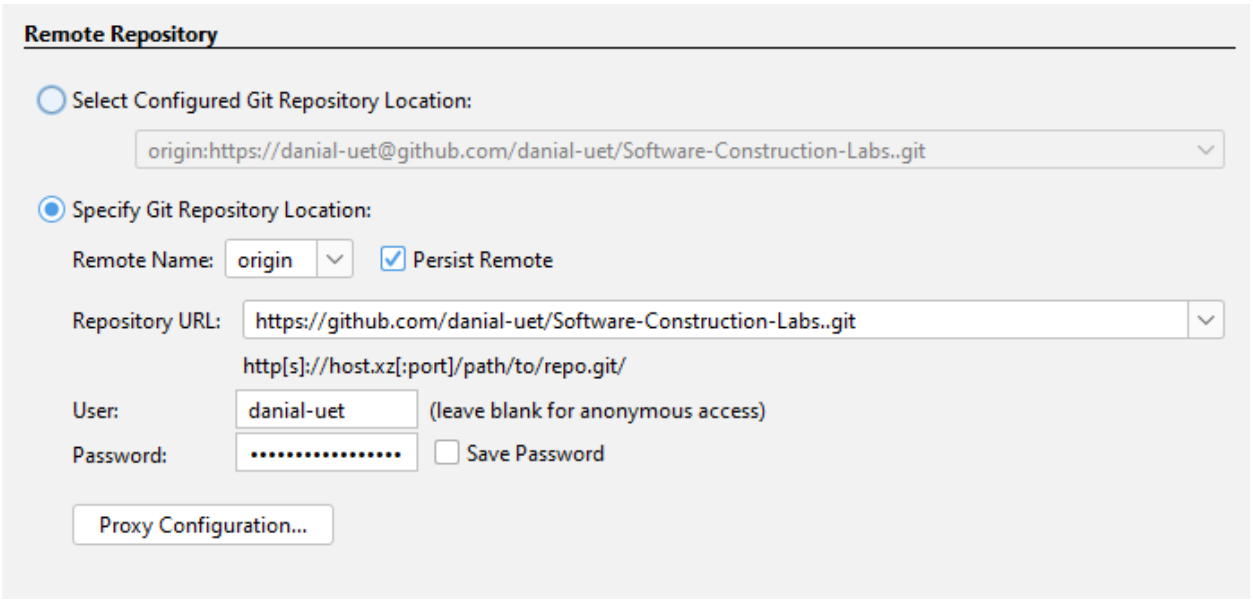


Output

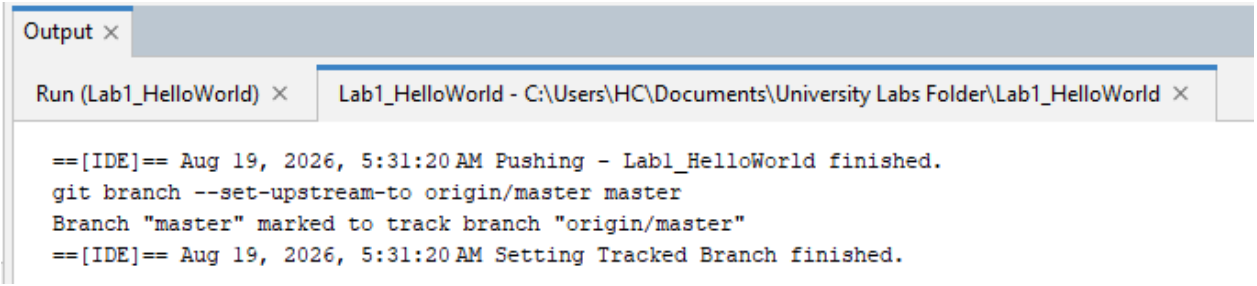


7. Right-click your project name, go to Git > Remote > Push.
   - Paste the repository URL from GitHub into the Remote Repository URI field.
   - Enter your GitHub Username and Personal Access Token (or password).
   - Select the master or main branch and click Finish.





Output



Result Screenshots:

